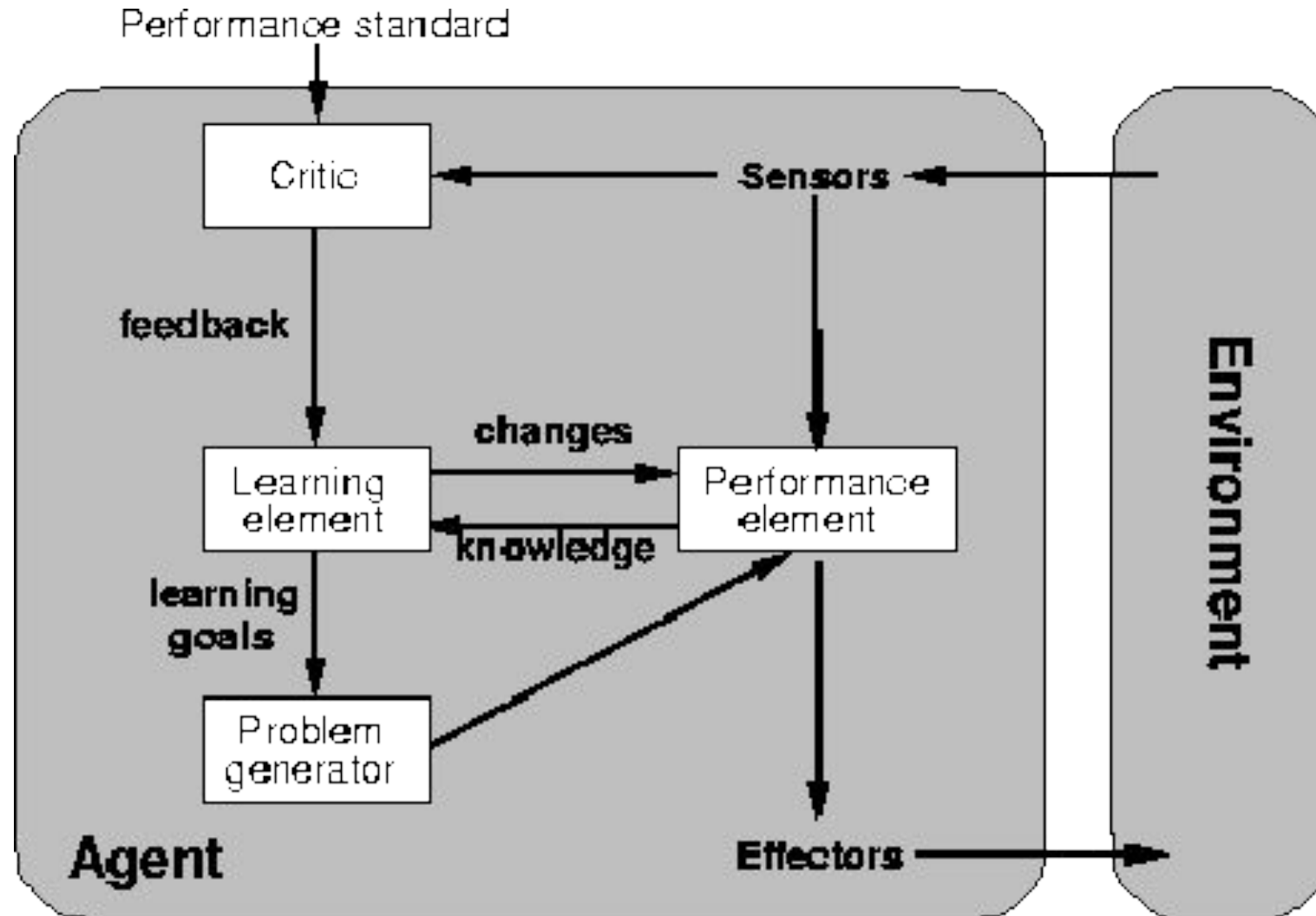


CSY3025 Artificial Intelligence Techniques

Deep Learning

A general model of learning agents



What is Machine Learning?

“Learning is any process by which a system improves performance from experience.”

- Herbert Simon

Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

- improve their performance P

- at some task T

- with experience E .

A well-defined learning task is given by $\langle P, T, E \rangle$.

Defining the Learning Task

Improve on task T, with respect to
performance metric P, based on experience E

T: Playing checkers

P: Percentage of games won against an arbitrary opponent

E: Playing practice games against itself

T: Recognizing hand-written words

P: Percentage of words correctly classified

E: Database of human-labeled images of handwritten words

T: Driving on four-lane highways using vision sensors

P: Average distance traveled before a human-judged error

E: A sequence of images and steering commands recorded while
observing a human driver.

T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

Deep Learning



- Virtual assistants

- Whether it's Alexa or Siri or Cortana, the virtual assistants of online service providers use deep learning to help understand your speech and the language humans use when they interact with them.

- Translations

- In a similar way, deep learning algorithms can automatically translate between languages. This can be powerful for travelers, business people and those in government.

- Vision for driverless delivery trucks, drones and autonomous cars

- The way an autonomous vehicle understands the realities of the road and how to respond to them whether it's a stop sign, a ball in the street or another vehicle is through deep learning algorithms.

- Chatbots and service bots

- Chatbots and service bots that provide customer service for a lot of companies are able to respond in an intelligent and helpful way to an increasing amount of auditory and text questions thanks to deep learning.

- Facial recognition

- Deep learning is being used for facial recognition not only for security purposes but for tagging people on Facebook posts and we might be able to [pay items in a store just by using our faces](#) in the near future.

- Medicine and pharmaceuticals

- From disease and tumor diagnoses to personalized medicines created specifically for an individual's genome, deep learning in the medical field has the attention of many of the largest pharmaceutical and medical companies.

- Personalized shopping and entertainment

- Ever wonder where Amazon comes up with ideas for what you should buy next and those suggestions are exactly what you need but just never knew it before? Yep, it's deep-learning algorithms at work.

Some more examples of tasks that are best solved by using a learning algorithm

- Recognizing patterns:
 - _ Facial identities or facial expressions
 - _ Handwritten or spoken words
 - _ Medical images
- Generating patterns:
 - _ Generating images or motion sequences
- Recognizing anomalies:
 - _ Unusual credit card transactions
 - _ Unusual patterns of sensor readings in a nuclear power plant
- Prediction:
 - _ Future stock prices or currency exchange rates

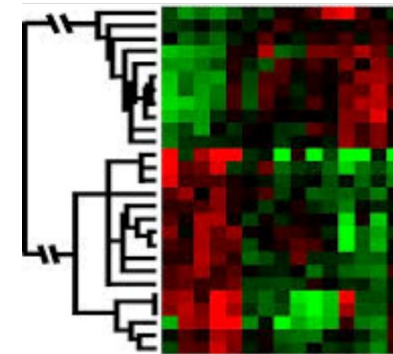
When Would We Use Machine Learning?

- When patterns exist in our data
 - Even if we don't know what they are
 - Or perhaps especially when we don't know what they are
- We can not pin down the functional relationships mathematically
 - Else we would just code up the algorithm
- When we have lots of (unlabeled) data
 - Labeled training sets harder to come by
 - Data is of high-dimension
 - High dimension “features”
 - For example, sensor data
 - Want to “discover” lower-dimension representations
 - Dimension reduction
- Aside: Machine Learning is heavily focused on implementability
 - Frequently using well known numerical optimization techniques
 - Lots of open source code available
 - See e.g., libsvm (Support Vector Machines): <http://www.csie.ntu.edu.tw/~cjlin/libsvm/>
 - Most of my code in python: <http://scikit-learn.org/stable/> (many others)
 - Languages (e.g., octave: <https://www.gnu.org/software/octave/>)

When Do We Use Machine Learning?

ML is used when:

- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



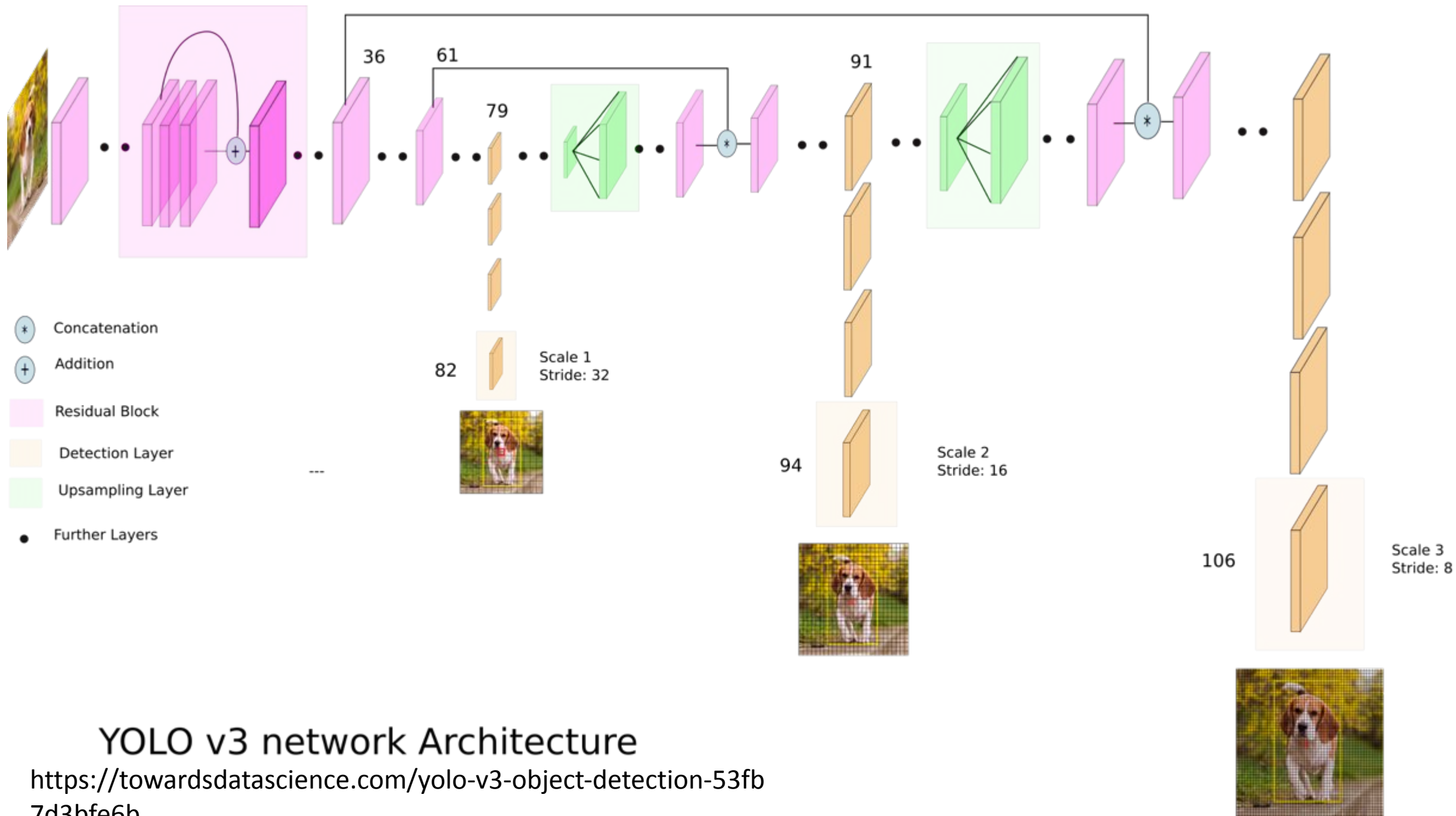
Learning isn't always useful:

- There is no need to “learn” to calculate payroll

Sample Applications

- Web search
- Computational biology
- Finance
- E-commerce
- Space exploration
- Robotics
- Information extraction
- Social networks
- Debugging software
- [Your favorite area]





YOLO v3 network Architecture

<https://towardsdatascience.com/yolo-v3-object-detection-53fb7d3bfe6b>



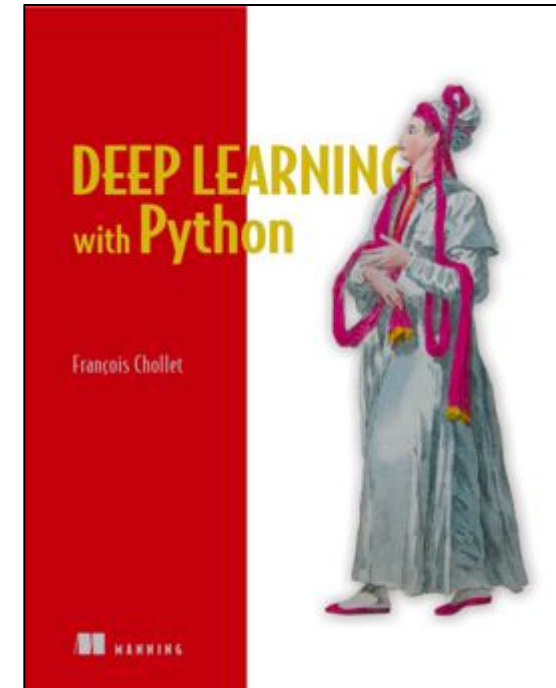
...

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data/dataset/logo32plus/images/cocacola/002076.jpg	649,192,708,219,7		
data/dataset/logo32plus/images/cocacola/002077.jpg	404,240,516,351,7		
data/dataset/logo32plus/images/cocacola/002078.jpg	468,261,529,315,7		
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data/dataset/logo32plus/images/cocacola/002082.jpg	256,576,476,671,7		
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data/dataset/logo32plus/images/cocacola/002084.jpg	319,132,761,267,7		
data/dataset/logo32plus/images/cocacola/002085.jpg	648,630,680,663,7		
data/dataset/logo32plus/images/cocacola/002086.jpg	269,139,370,177,7	346,475,402,497,7	528,398,595,432,7
586,308,623,329,7			
data/dataset/logo32plus/images/cocacola/002087.jpg	110,425,297,775,7		
data/dataset/logo32plus/images/cocacola/002088.jpg	666,419,726,459,7		
data/dataset/logo32plus/images/cocacola/002089.jpg	103,326,283,392,7		
data/dataset/logo32plus/images/cocacola/002090.jpg	181,568,481,749,7		
data/dataset/logo32plus/images/cocacola/002092.jpg	660,202,769,440,7	830,248,884,310,7	147,346,262,381,7
data/dataset/logo32plus/images/cocacola/002093.jpg	216,398,519,475,7		
data/dataset/logo32plus/images/cocacola/002094.jpg	380,334,645,450,7		
data/dataset/logo32plus/images/cocacola/002095.jpg	533,724,679,782,7	112,479,209,529,7	
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data/dataset/logo32plus/images/cocacola/002098.jpg	397,304,611,412,7		
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data/dataset/logo32plus/images/cocacola/002103.jpg	543,556,684,681,7		

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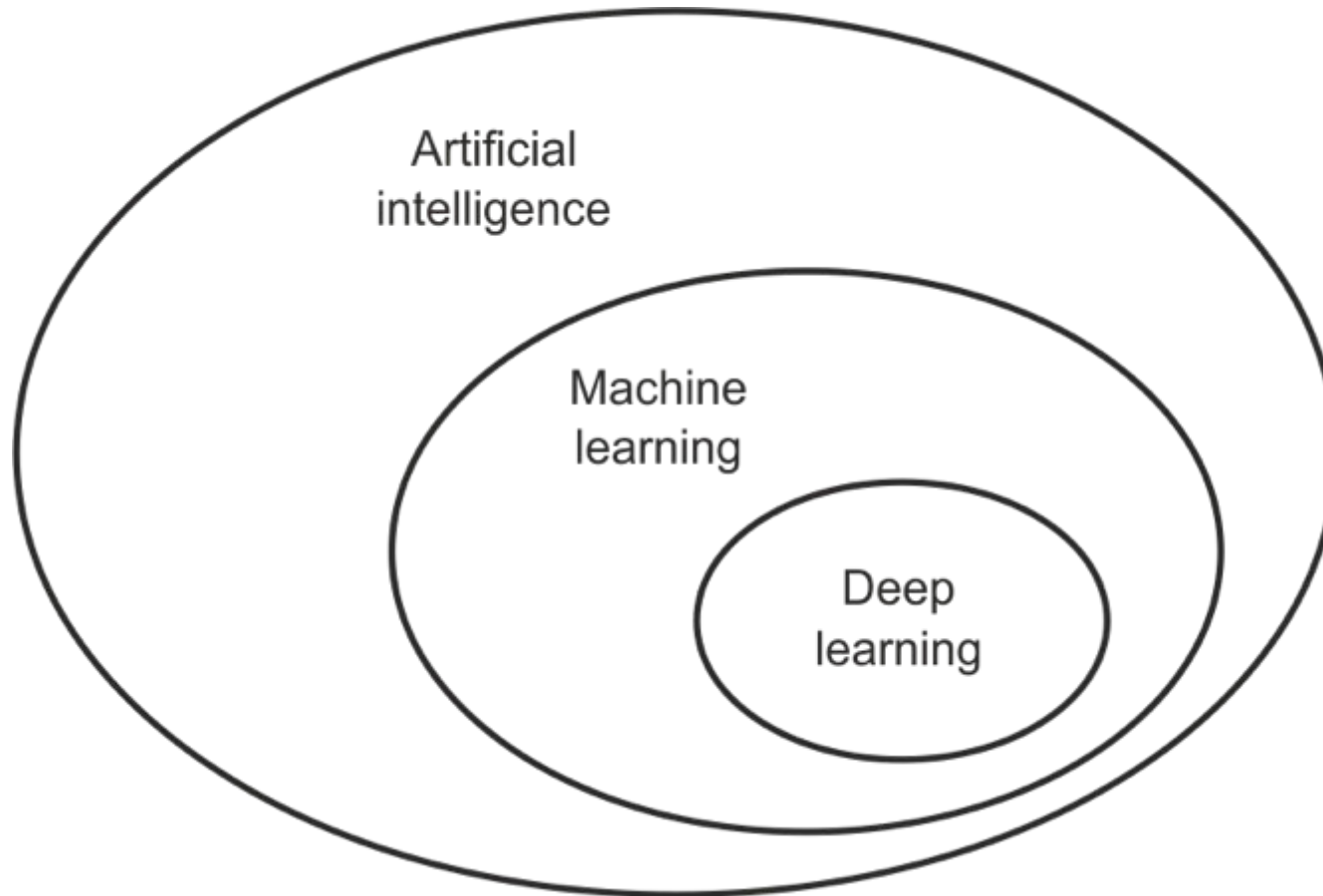
Reference book

- **Deep learning with Python** by Francois Chollet
- A practical, hands-on exploration of deep learning.
- Python deep-learning framework *Keras*, with *TensorFlow* as a backend engine.
- You'll be able to tackle *real-world problems*: image classification, timeseries forecasting, sentiment analysis, image and text generation, and more.



What is deep learning

What are artificial intelligence, machine learning, and deep learning?

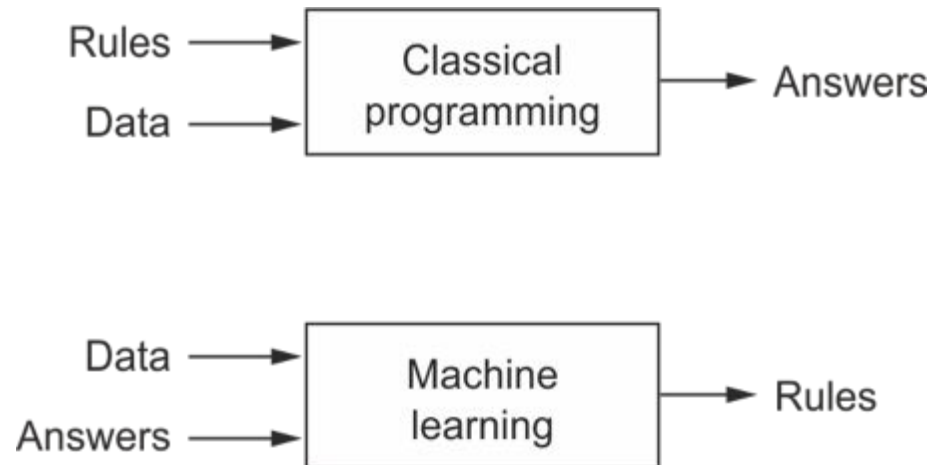


Artificial intelligence

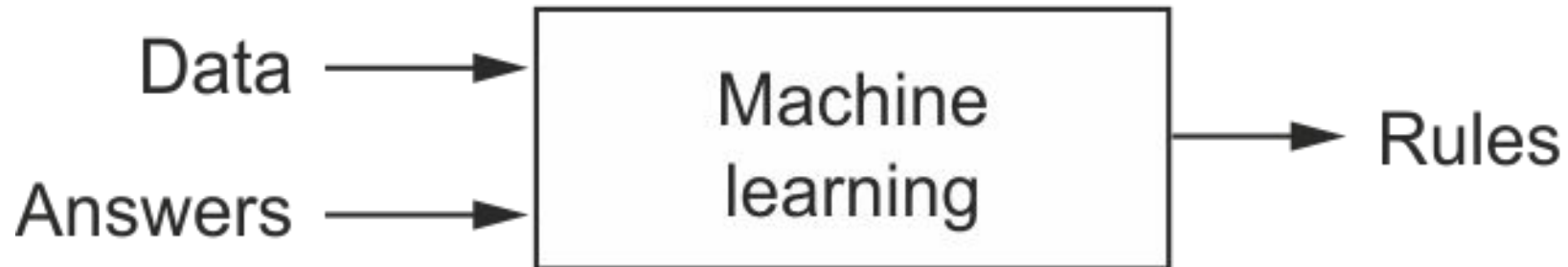
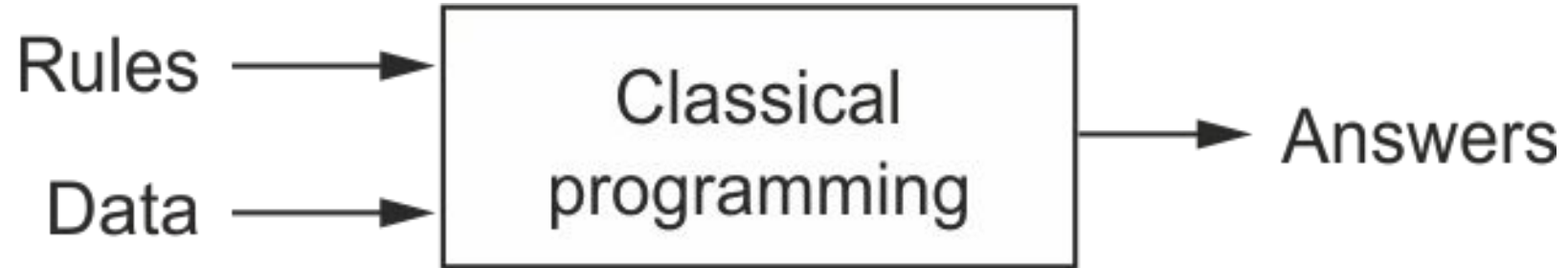
- Artificial intelligence was born in the 1950s, when a handful of pioneers in field of computer science started asking whether computers could be made to “think” — we’re still exploring today.
- A concise definition of the field would be as follows: *the effort to automate intellectual tasks normally performed by humans*.
- As such, AI is a general field that encompasses machine learning and deep learning, but that also includes many more approaches that don’t involve any learning.
 - Early chess programs, for instance, only involved hardcoded rules crafted by programmers, and didn’t qualify as machine learning. For a fairly long time, many experts believed that human-level artificial intelligence could be achieved by having programmers handcraft a sufficiently large set of explicit rules for manipulating knowledge. This approach is known as *symbolic AI*, and it was the dominant paradigm in AI from the 1950s to the late 1980s. It reached its peak popularity during the *expert systems* boom of the 1980s.
- Although symbolic AI proved suitable to solve well-defined, logical problems, such as playing chess, it turned out to be intractable to figure out explicit rules for solving more complex, fuzzy problems, such as image classification, speech recognition, and language translation. A new approach arose to take symbolic AI’s place: *machine learning*.

Machine Learning

- Machine learning arises from this question: could a computer go beyond “what we know how to order it to perform” and learn on its own how to perform a specified task? This question opens the door to a new programming paradigm.
 - In classical programming, the paradigm of symbolic AI, humans input rules (a program) and data to be processed according to these rules, and out come answers.
 - With machine learning, humans input data as well as the answers expected from the data, and out come the rules. These rules can then be applied to new data to produce original answers.



Machine Learning



Learning representations from data

- ***Input data points***— For instance, if the task is speech recognition, these data points could be sound files of people speaking. If the task is image tagging, they could be pictures.
- ***Examples of the expected output***— In a speech-recognition task, these could be human-generated transcripts of sound files. In an image task, expected outputs could be tags such as “dog,” “cat,” and so on.
- ***A way to measure whether the algorithm is doing a good job***— This is necessary in order to determine the distance between the algorithm’s current output and its expected output. The measurement is used as a feedback signal to adjust the way the algorithm works. This adjustment step is what we call *learning*.

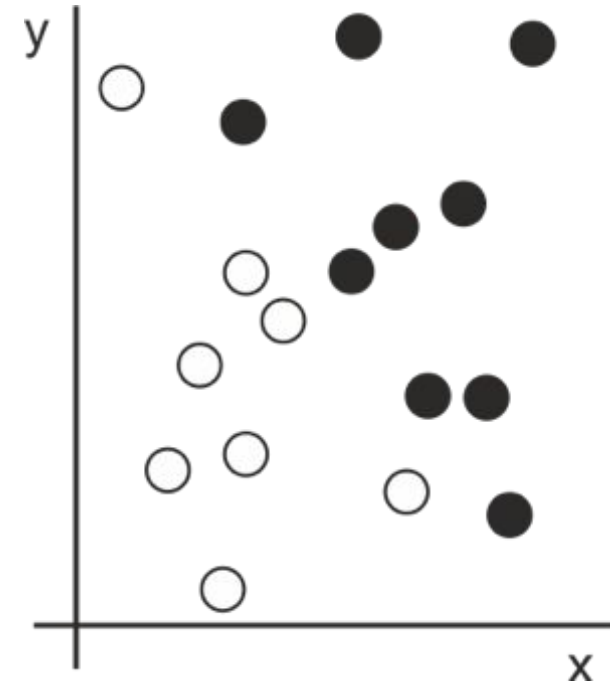
Learning representations from data

- A machine-learning model transforms its input data into meaningful outputs, a process that is “learned” from exposure to known examples of inputs and outputs.
- Therefore, the central problem in machine learning and deep learning is to *meaningfully transform data*: in other words, to learn useful *representations* of the input data at hand—representations that get us closer to the expected output.
- Machine-learning models are all about finding appropriate representations for their input data—transformations of the data that make it more amenable to the task at hand, such as a classification task.



Learning representations from data

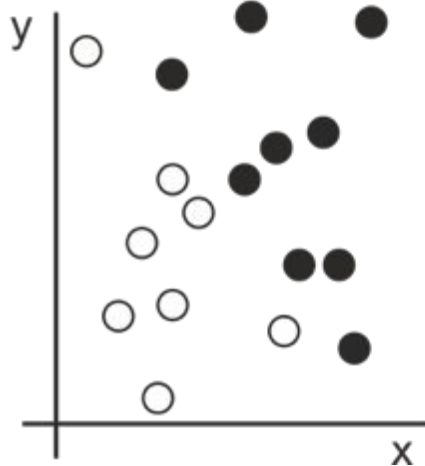
- Let's make this concrete. Consider an x-axis, a y-axis, and some points represented by their coordinates in the (x, y) system. We want to develop an algorithm that can take the coordinates (x, y) of a point and output whether that point is likely to be black or to be white.



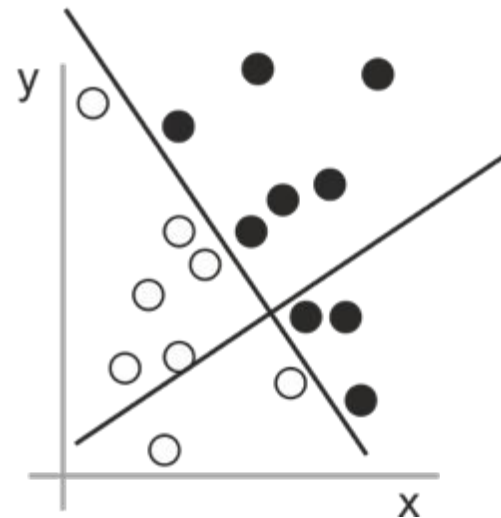
Learning representations from data

- What we need here is a new representation of our data that cleanly separates the white points from the black points. One transformation we could use would be a coordinate change.
- In this new coordinate system, the coordinates of our points can be said to be a new representation of our data. And it's a good one! With this representation, the black/white classification problem can be expressed as a simple rule: "Black points are such that $x > 0$," or "White points are such that $x < 0$." This new representation basically solves the classification problem.

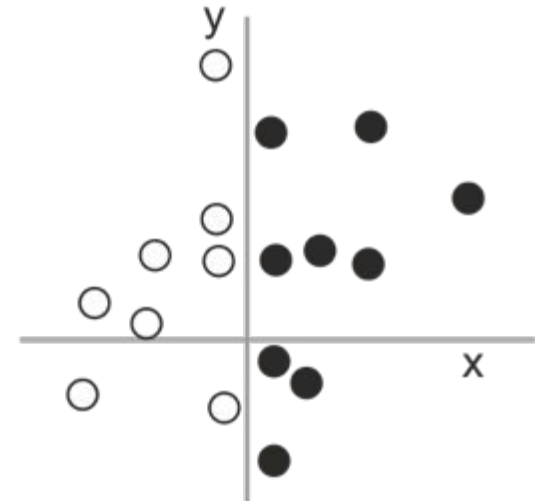
1: Raw data



2: Coordinate change



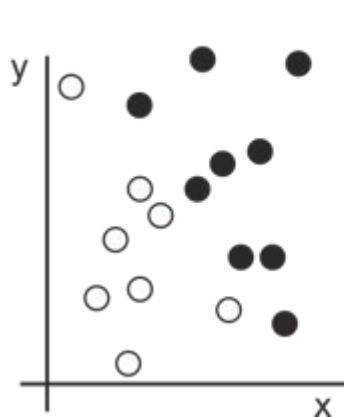
3: Better representation



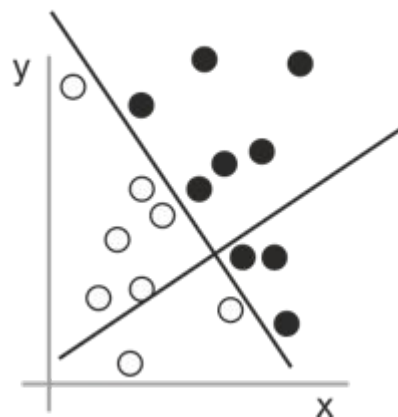
Learning representations from data

- All machine-learning algorithms consist of automatically finding such transformations that turn data into more-useful representations for a given task. These operations can be coordinate changes, linear projections, translations, nonlinear operations, and so on. Machine-learning algorithms aren't usually creative in finding these transformations; they're merely searching through a predefined set of operations, called a *hypothesis space*.
- So that's what machine learning is, technically: searching for useful representations of some input data, within a predefined space of possibilities, using guidance from a feedback signal. This simple idea allows for solving a remarkably broad range of intellectual tasks, from speech recognition to autonomous car driving.

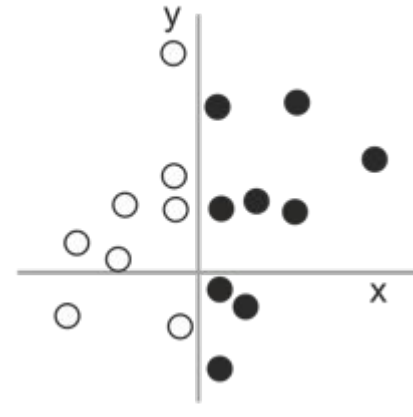
1: Raw data



2: Coordinate change

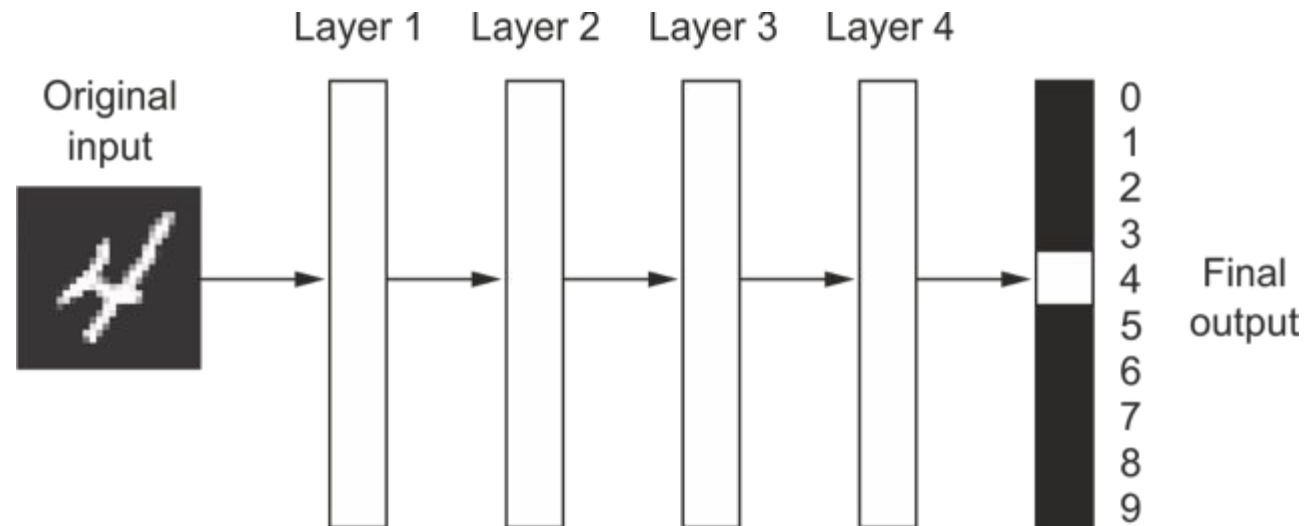


3: Better representation



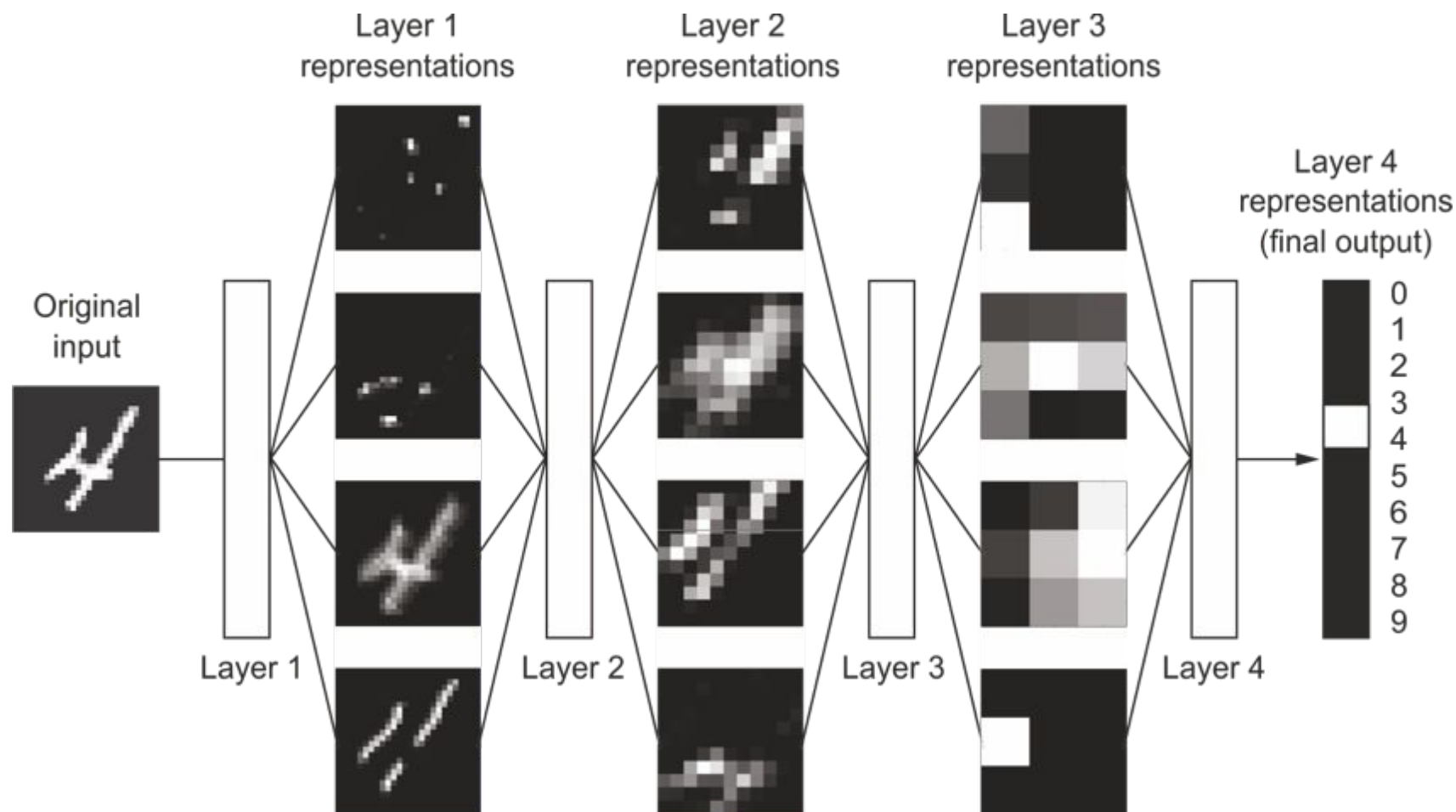
The “deep” in deep learning

- Deep learning is a new take on learning representations from data that puts an emphasis on learning successive *layers* of increasingly meaningful representations.
 - How many layers contribute to a model of the data is called the *depth* of the model.
 - Modern deep learning often involves tens or even hundreds of successive layers of representations—and they’re all learned automatically from exposure to training data.
 - Meanwhile, other approaches to machine learning tend to focus on learning only one or two layers of representations of the data; hence, they’re sometimes called *shallow learning*.



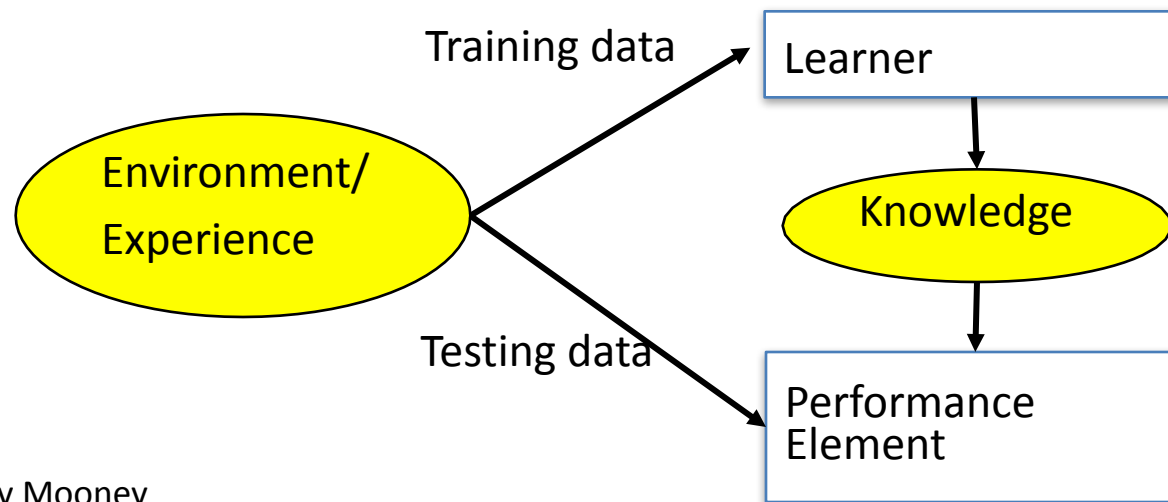
The “deep” in deep learning

- The network transforms the digit image into representations that are increasingly different from the original image and **increasingly informative** about the final result. You can think of a deep network as a multistage information-distillation operation, where information goes through successive filters and comes out increasingly *purified* (that is, useful with regard to some task).



Designing a Learning System

- Choose the training experience
- Choose exactly what is to be learned
 - i.e. the **target function**
- Choose how to represent the target function
- Choose a learning algorithm to infer the target function from the experience



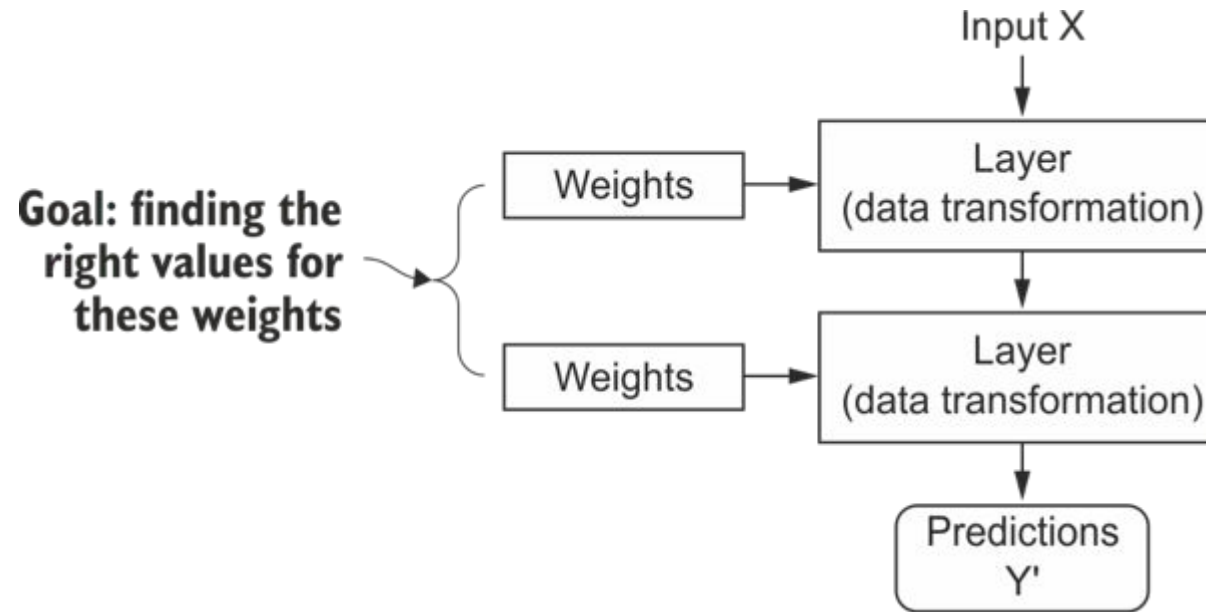
ML in a Nutshell

- Tens of thousands of machine learning algorithms
 - Hundreds new every year
- Every ML algorithm has three components:
 - **Representation**
 - **Optimization**
 - **Evaluation**

Training vs. Test Distribution

- We generally assume that the training and test examples are independently drawn from the same overall distribution of data
 - We call this “i.i.d” which stands for “independent and identically distributed”
- If examples are not independent, requires *collective classification*
- If test distribution is different, requires *transfer learning*

Understanding how deep learning works, in 3 figures - 1

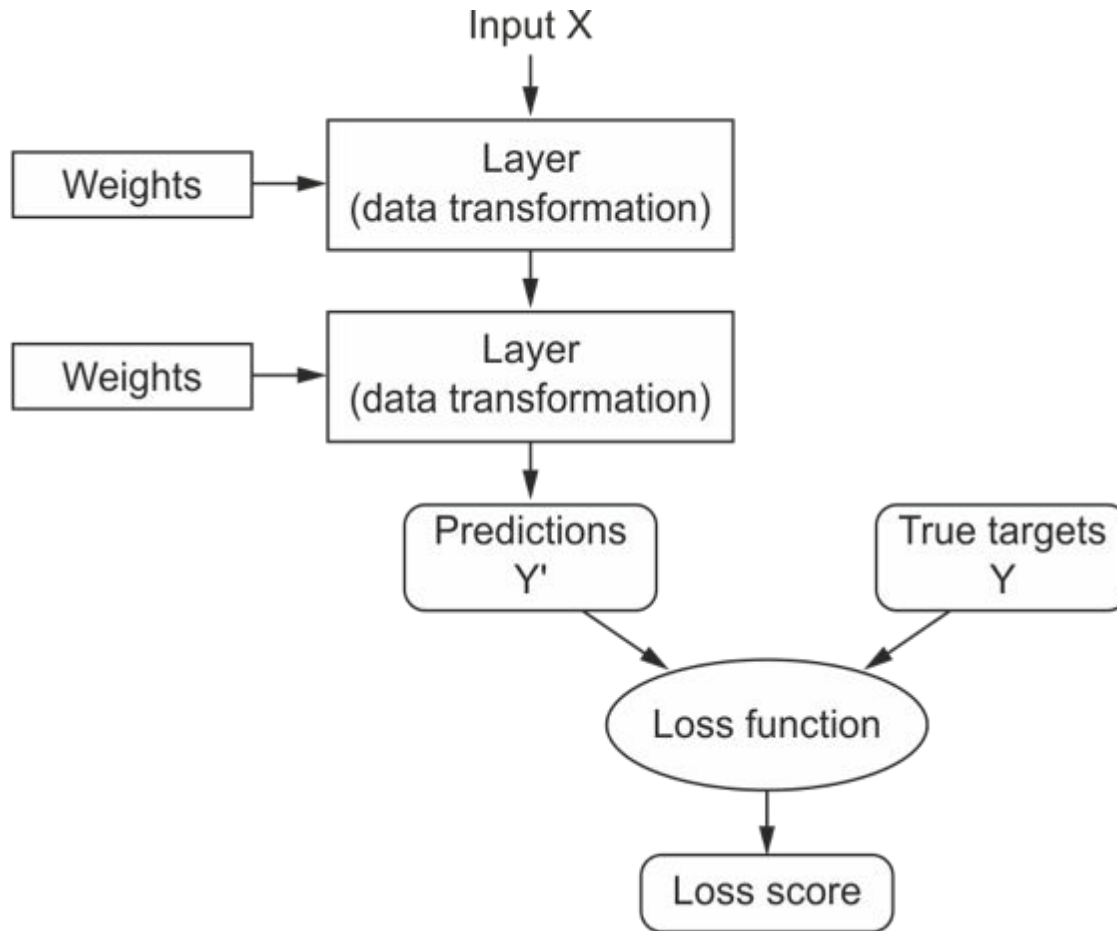


The specification of what a layer does to its input data is stored in the layer's weights, which in essence are a bunch of numbers.

In technical terms, we'd say that the transformation implemented by a layer is parameterized by its weights (parameters). In this context, learning means finding a set of values for the weights of all layers in a network, such that the network will correctly map example inputs to their associated targets.

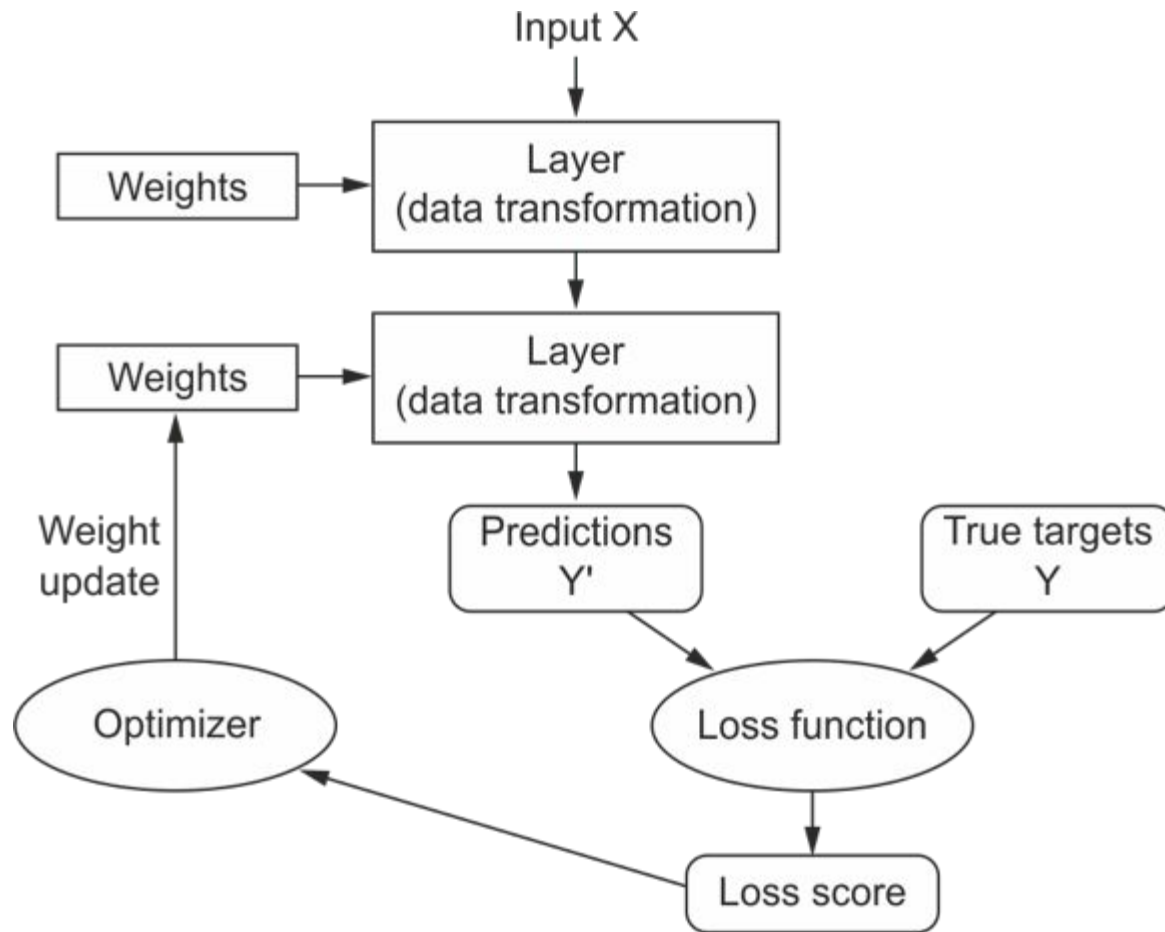
But here's the thing: a deep neural network can contain tens of millions of parameters. Finding the correct value for all of them may seem like a daunting task, especially given that modifying the value of one parameter will affect the behaviour of all the others!

Understanding how deep learning works, in 3 figures - 2



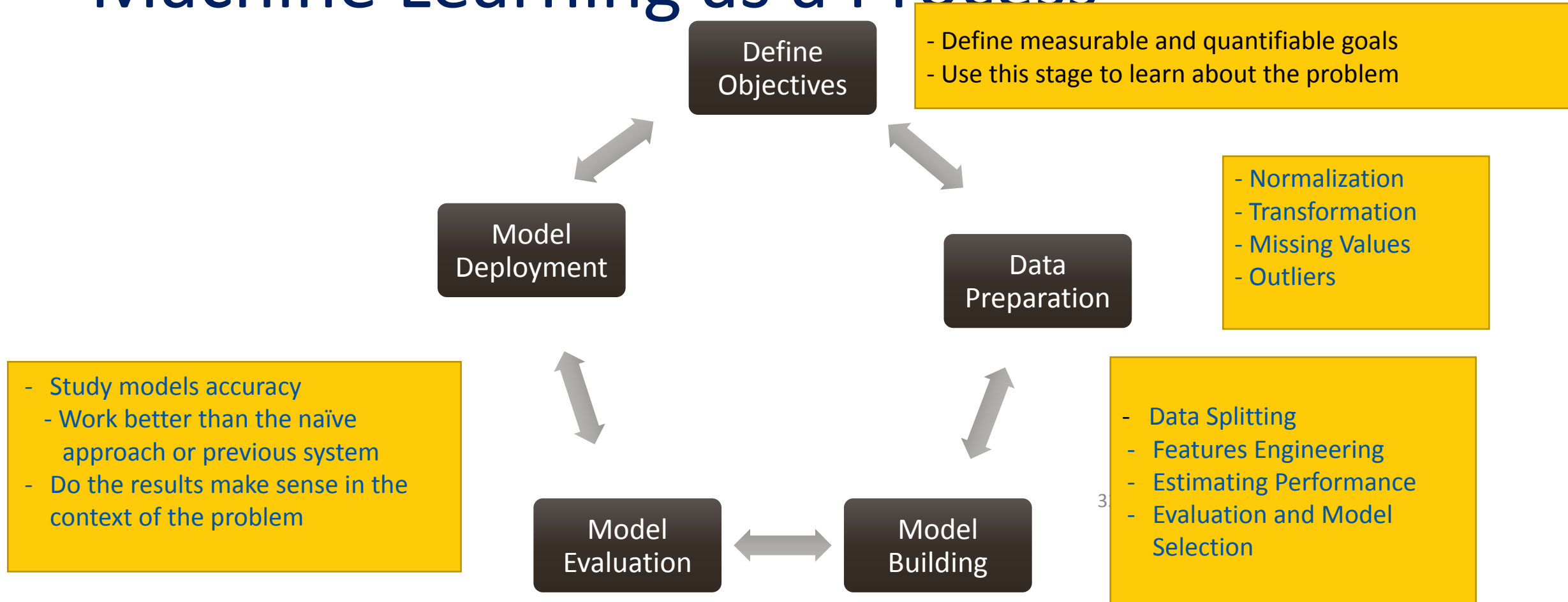
- To control something, first you need to be able to **observe** it.
- To control the output of a neural network, you need to be able to **measure** how far this output is from what you expected.
- This is the job of the ***loss function*** of the network, also called the ***objective function***.
 - The loss function takes the predictions of the network and the true target (what you wanted the network to output) and computes a distance score, capturing how well the network has done on this specific example.

Understanding how deep learning works, in 3 figures - 3



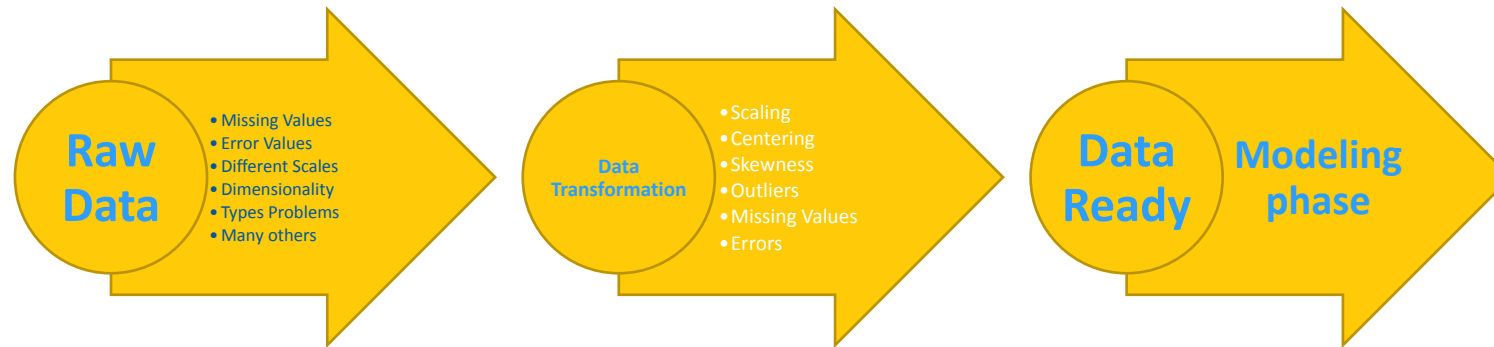
- The fundamental trick in deep learning is to use this score as a **feedback** signal to **adjust the value of the weights** a little, in a **direction** that will lower the loss score for the current example.
- This adjustment is the job of the **optimizer**, which implements what's called the **Backpropagation** algorithm: the central algorithm in deep learning.
 - Initially, the weights of the network are assigned random values, but with every example the network processes, the weights are adjusted a little in the correct direction, and the loss score decreases.
 - This is the training loop, which, repeated a sufficient number of times (typically tens of iterations over thousands of examples), yields weight values that minimize the loss function.
 - A network with a minimal loss is one for which the outputs are as close as they can be to the targets: a trained network. Once again, it's a simple mechanism that, once scaled, ends up looking like magic.

Machine Learning as a Process



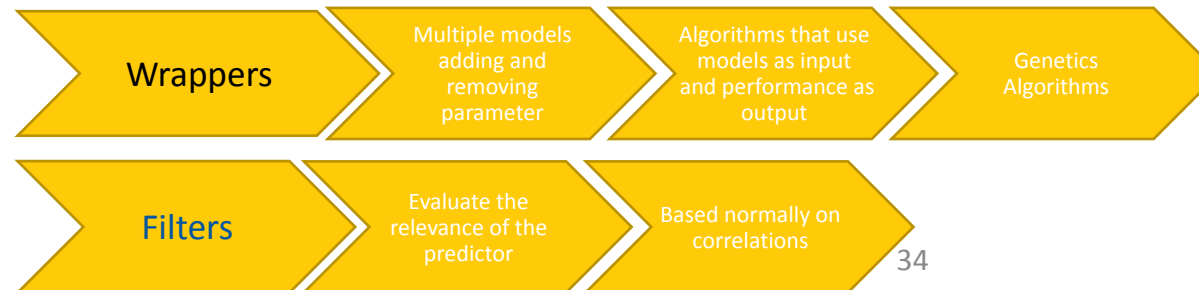
ML as a Process: Data Preparation

- Needed for several reasons
 - Some Models have strict data requirements
 - Scale of the data, data point intervals, etc
 - Some characteristics of the data may impact dramatically on the model performance
- Time on data preparation should not be underestimated



ML as a Process: Feature engineering

- Determine the predictors (features) to be used is one of the most critical questions
- Some times we need to add predictors
- **Reduce Number:**
 - Fewer predictors **more interpretable model and less costly**
 - Most of the models are **affected by high dimensionality, specially for non-informative predictors**



- Binning predictors

ML as a Process: Model Building

- Data Splitting
 - Allocate data to different tasks
 - model training
 - performance evaluation
 - Define Training, Validation and Test sets
- Feature Selection (Review the decision made previously)
- Estimating Performance
 - Visualization of results – discovery interesting areas of the problem space
 - Statistics and performance measures
- Evaluation and Model selection
 - The ‘no free lunch’ theorem no a priory assumptions can be made
 - Avoid use of favorite models if NEEDED

<http://playground.tensorflow.org/>



Epoch
000,533

Learning rate

0.03

Activation

Tanh

Regularization

None

Regularization rate

0

Problem type

Classification

DATA

Which dataset do you want to use?



Ratio of training to test data: 50%



Noise: 0



Batch size: 10



REGENERATE

FEATURES

Which properties do you want to feed in?



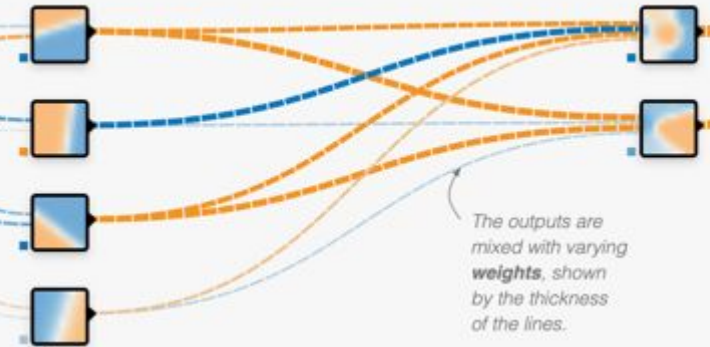
+ - 2 HIDDEN LAYERS

+ -

4 neurons

+ -

2 neurons



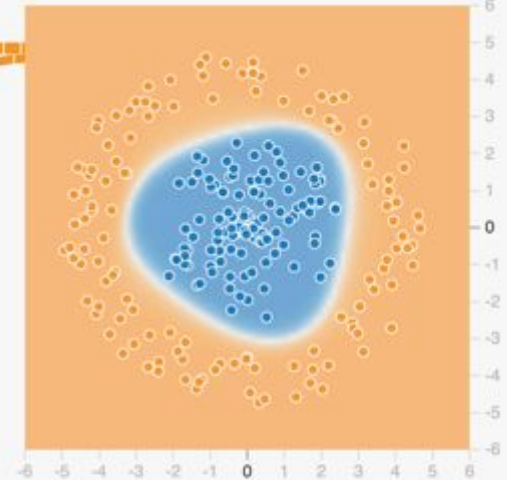
This is the output from one **neuron**. Hover to see it larger.

The outputs are mixed with varying **weights**, shown by the thickness of the lines.

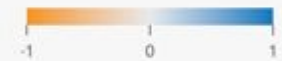
OUTPUT

Test loss 0.001

Training loss 0.001



Colors shows data, neuron and weight values.



☐ Show test data

☐ Discretize output

Various Function Representations

- Numerical functions
 - Linear regression
 - Neural networks
 - Support vector machines
- Symbolic functions
 - Decision trees
 - Rules in propositional logic
 - Rules in first-order predicate logic
- Instance-based functions
 - Nearest-neighbor
 - Case-based
- Probabilistic Graphical Models
 - Naïve Bayes
 - Bayesian networks
 - Hidden-Markov Models (HMMs)
 - Probabilistic Context Free Grammars (PCFGs)
 - Markov networks

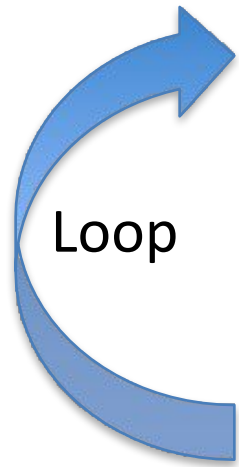
Various Search/Optimization

- Algorithms
 - Gradient descent
 - Perceptron
 - Backpropagation
 - Dynamic Programming
 - HMM Learning
 - PCFG Learning
 - Divide and Conquer
 - Decision tree induction
 - Rule learning
 - Evolutionary Computation
 - Genetic Algorithms (GAs)
 - Genetic Programming (GP)
 - Neuro-evolution

Evaluation

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
-
- Margin
- Entropy
-
- K-L divergence
- etc.

ML in Practice



- Understand domain, prior knowledge, and goals
- Data integration, selection, cleaning, pre-processing, etc.
- Learn models
- Interpret results
- Consolidate and deploy discovered knowledge

History of Machine Learning

- 1950s
 - Samuel's checker player
 - Selfridge's Pandemonium
- 1960s:
 - Neural networks: Perceptron
 - Pattern recognition
 - Learning in the limit theory
 - Minsky and Papert prove limitations of Perceptron
- 1970s:
 - Symbolic concept induction
 - Winston's arch learner
 - Expert systems and the knowledge acquisition bottleneck
 - Quinlan's ID3
 - Michalski's AQ and soybean diagnosis
 - Scientific discovery with BACON
 - Mathematical discovery with AM

History of Machine Learning (cont.)

- 1980s:
 - Advanced decision tree and rule learning
 - Explanation-based Learning (EBL)
 - Learning and planning and problem solving
 - Utility problem
 - Analogy
 - Cognitive architectures
 - Resurgence of neural networks (connectionism, backpropagation)
 - Valiant's PAC Learning Theory
 - Focus on experimental methodology
- 1990s
 - Data mining
 - Adaptive software agents and web applications
 - Text learning
 - Reinforcement learning (RL) Inductive Logic Programming (ILP) Ensembles: Bagging, Boosting, and Stacking Bayes Net learning

History of Machine Learning (cont.)

- 2000s
 - _ Support vector machines & kernel methods
 - _ Graphical models
 - _ Statistical relational learning
 - _ Transfer learning
 - _ Sequence labeling
 - _ Collective classification and structured outputs
 - _ Computer Systems Applications (Compilers, Debugging, Graphics, Security)
 - _ E-mail management
 - _ Personalized assistants that learn
- 2010s
 - _ Learning in robotics and vision
 - _ Deep learning systems
 - _ Learning for big data
 - _ Bayesian methods
 - _ Multi-task & lifelong learning
 - _ Applications to vision, speech, social networks, learning to read, etc.
 - _ ???

TODO

- Google Colab - <https://colab.research.google.com/>